

Walmart Sales Data Analysis

Final Business Questions – Python EDA and MySQL Workbench

Part 1: Exploratory Data Analysis (EDA) – 21 Questions

1. Data Overview and Data Quality Check

Business Question: What are the structure, data types, missing values, duplicates, and basic statistics of the dataset?

2. Category-wise Revenue Analysis

Business Question: How much revenue is generated by each product category?

3. Top Cities by Revenue

Business Question: Which cities generate the highest revenue?

4. Payment Method-wise Revenue

Business Question: How much revenue is generated through each payment method?

5. Category-wise Average Rating

Business Question: Which categories receive the highest and lowest average customer ratings?

6. Category-wise Average Profit Margin

Business Question: Which categories have the highest and lowest average profit margin?

7. Best Performing Branch

Business Question: Which branch performs best based on total revenue?

8. Overall Business Performance (KPI Summary)

Business Question: What are the overall business KPIs such as total revenue, transactions, average order value, and average rating?

9. Top Categories by Revenue

Business Question: Which categories are the top revenue contributors?

10. Best and Worst Performing Category

Business Question: Which category performs best and which performs worst based on revenue?

11. Branch Performance Analysis

Business Question: How do branches compare based on revenue, transactions, ratings, and quantity sold?

12. City Performance Analysis

Business Question: How do cities compare based on revenue, transactions, and customer ratings?

13. Revenue vs Profit Analysis

Business Question: How does revenue compare with profit across categories?

14. High-Value Transactions Analysis

Business Question: Which transactions are high-value based on the selected sales threshold?

15. High-Value Transaction Revenue by Category

Business Question: Which categories contribute the most revenue from high-value transactions?

16. Rating vs Sales Analysis

Business Question: Is there a visible relationship between customer ratings and sales?

17. Payment Method Analysis

Business Question: How do payment methods compare based on transactions, revenue, and average order value?

18. Category × City Analysis

Business Question: How does category performance vary across different cities?

19. Category-wise Sales Across Top 15 Cities

Business Question: How does category-wise sales performance vary across the top 15 cities?

20. Profit Analysis

Business Question: How is profit distributed across the business data?

21. Most and Least Profitable Category

Business Question: Which categories generate the highest and lowest total profit?

Part 2: SQL Business Questions – 24 Questions

1. Total Revenue

Business Question: What is the total revenue generated by Walmart?

2. Total Transactions

Business Question: How many total transactions are recorded in the dataset?

3. Unique Cities

Business Question: How many unique cities are present in the dataset?

4. Top 10 High-Value Transactions

Business Question: Which are the top 10 transactions with the highest total sales?

5. Category-wise Revenue

Business Question: What is the total revenue generated by each product category?

6. City-wise Revenue

Business Question: Which cities generate the highest total revenue?

7. Average Transaction Value

Business Question: What is the average sales value per transaction?

8. Category-wise Average Rating

Business Question: What is the average customer rating for each category?

9. Branch-wise Revenue

Business Question: What is the total revenue generated by each branch?

10. Categories with High Revenue

Business Question: Which categories have generated revenue greater than the selected revenue threshold?

11. City-wise Transaction Count

Business Question: How many transactions were made in each city?

12. Transactions Above Average Sales

Business Question: Which transactions have total sales greater than the average transaction value?

13. Best Performing Category

Business Question: Which category generates the highest total revenue?

14. Highest Revenue City

Business Question: Which city generated the highest revenue?

15. Rank Categories by Revenue

Business Question: How do categories rank based on their total revenue?

16. Rank Branches by Revenue

Business Question: How do branches rank based on their total revenue?

17. Running Total of Sales

Business Question: How does the cumulative total sales value grow over time?

18. Daily Sales Analysis

Business Question: What is the total revenue generated on each date?

19. Monthly Sales Analysis

Business Question: How does total revenue vary month-wise?

20. Best Sales Day

Business Question: Which date generated the highest total revenue?

21. CTE – Best Performing Category

Business Question: Using a CTE, calculate category-wise revenue and find the best-performing category.

22. CTE – Cities Above Average Revenue

Business Question: Using a CTE, find cities whose revenue is above the average city revenue.

23. Create a Sales Summary View

Business Question: Create a reusable virtual table showing category-wise total revenue and total profit.

24. Find Best Category Using View

Business Question: Using the sales summary view, which category has the highest total revenue?